

FACULTY OF ENGINEERING AND TECHNOLOGY
SCHOOL OF COMPUTING AND ARTIFICIAL INTELLIGENCE

Capstone Project Logbook

Student Information

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Student ID:	23076003
Programme:	Bachelor of Computer Science (Hons)
Project Title:	Evidence-Grounded Local-LLM Explanations for Credit Card Fraud Alert Review
Supervisor:	Dr Tang Tiong Yew

Weekly Entry Template

Week Number:	1
Date Range:	11-17 May 2026

Week 1: Project Restart and CP1-to-CP2 Scope Review

Objectives for the Week: Restart the project from the approved CP1 proposal and identify what had to be delivered during CP2.

Activities Completed: I reviewed the CP1 proposal, including the original Autoencoder-XGBoost detector, SHAP explanations, local LLM, privacy rationale, and guardrail plan. I separated the work into two questions: whether the hybrid detector improved fraud detection, and whether generated explanations could be kept faithful to model evidence. This prevented the LLM from being treated as a second fraud detector.

Evidence/files: CP1/01_FINAL_SUBMISSION/Proposal_Capstone_Project.pdf; final report Sections 1.2-1.5 and 3.1.

Challenges Encountered (if any): The proposal combined model development, explainable AI, an LLM, and an application. Without a clear hierarchy, CP2 could have looked like several unrelated projects.

Solutions / Actions Taken (if any): I separated the work into a detector-performance question and an explanation-faithfulness question, with the detector as the prediction source, SHAP as evidence, and the LLM only as an articulation layer.

Progress Reflection: I decided that the detector would remain the source of the prediction, SHAP would provide attribution evidence, and the LLM would only articulate that evidence. This role separation later became the central project design.

Plans for Next Week: Confirm project continuity and map the semester deliverables.

Supervisor's Feedback (if any): —

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Week Number:	2
Date Range:	18-24 May 2026

Week 2: Supervisor Continuity and Semester Planning

Objectives for the Week: Confirm the CP2 administrative starting point and plan the work against the Week 12 submission deadline.

Activities Completed: I confirmed that Dr Claymond Lim Wei Xiang remained the supervisor from CP1. I mapped the project work to the CP2 teaching timeline: design and preparation first, implementation and evaluation next, followed by the final report, logbook, presentation, and system demonstration. I also kept the project title open to revision if the detector results did not support the original Autoencoder emphasis.

Evidence/files: docs/references/CP2_Timeline_2026.png; CP1 proposal cover; final report cover and Section 1.3.

Challenges Encountered (if any): A schedule based only on model training would leave insufficient time for explanation validation, application work, writing, and demonstration preparation.

Solutions / Actions Taken (if any): I mapped design, implementation, evaluation, writing, and demonstration tasks to the official CP2 weeks, while leaving the title open to evidence-driven revision.

Progress Reflection: Treating the report, reproducibility evidence, and system demonstration as project outputs from the beginning reduced the risk of building a model that could not later be explained or defended.

Plans for Next Week: Revisit the literature and identify the most defensible research gap.

Supervisor's Feedback (if any): —

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Week Number:	3
Date Range:	25-31 May 2026

Week 3: Literature Review and Contribution Reframing

Objectives for the Week: Reassess which part of the proposed project could make a credible FYP contribution.

Activities Completed: I reviewed literature on imbalanced fraud detection, SMOTE, cost-sensitive learning, autoencoder features, XGBoost, SHAP, narrative explanations, local language models, hallucination, and programmable guardrails. Existing work already covered hybrid fraud detectors and LLM-based explanation. The more specific gap was the measurement and fail-closed control of LLM text against fixed SHAP evidence.

***Evidence/files:** The starting literature foundation is recorded in the CP1 literature review. The resulting CP2 synthesis was later formalised in final report Chapter 2 and Table 2.1 and verified in docs/reviews/2026-07-26-final-reference-audit.md.*

Challenges Encountered (if any): Claiming that Autoencoder plus XGBoost or “using an LLM for explanations” was new would be difficult to defend against prior research.

Solutions / Actions Taken (if any): I reframed the contribution around local generation, minimised evidence, deterministic validation, measured failure rates, and fail-closed fallback.

Progress Reflection: I reframed the intended contribution as an evaluated explanation-delivery boundary: local generation, minimized evidence, code-level validation, measured failure rates, and deterministic fallback. The detector benchmark would test the original CP1 assumption rather than guarantee it.

Plans for Next Week: Define the detector comparison, metrics, and research questions before implementation.

Supervisor's Feedback (if any): —

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Week Number:	4
Date Range:	1-7 June 2026

Week 4: Experimental Groups and Evaluation Design

Objectives for the Week: Design a fair detector experiment that could test the CP1 hybrid-model assumption.

Activities Completed: I specified six detector groups: baseline XGBoost, SMOTE-XGBoost, reconstruction-error hybrid, reconstruction-error plus SMOTE, cost-sensitive XGBoost, and latent-feature hybrid. Five fixed seeds were planned for every group. Average Precision was selected as the primary ranking metric, with precision, recall, F1, ROC-AUC, ranked-alert metrics, confusion counts, and runtime retained for interpretation.

***Evidence/files:** This design was later formalised in final report Sections 3.4-3.5 and Tables 3.1-3.2 and implemented in configs/g0_xgb.yaml to configs/g7_ae_latent_xgb.yaml.*

Challenges Encountered (if any): Fraud accuracy is misleading because non-fraud cases dominate the dataset. A single metric could also hide operational trade-offs between missed frauds and unnecessary reviews.

Solutions / Actions Taken (if any): I fixed the six experimental groups, five seeds, shared split, and multi-metric evaluation before training so the original hybrid hypothesis could fail fairly.

Progress Reflection: The multi-group, multi-seed design allowed the original hybrid idea to fail without invalidating the project. It also created a stronger basis for choosing one frozen detector for later SHAP and narrative work.

Plans for Next Week: Define the data split, preprocessing, identity, and leakage controls.

Supervisor's Feedback (if any): —

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Week Number:	5
Date Range:	8-14 June 2026

Week 5: Dataset and Leakage-Control Design

Objectives for the Week: Establish a reproducible data protocol before model training.

Activities Completed: I retained the European Credit Card Fraud dataset from the CP1 proposal and defined content deduplication, stable case identifiers, a stratified 70/15/15 train-validation-test split, train-only scaler fitting, and train-only SMOTE. Validation data would be used for tuning, detector selection, and threshold selection; test data would remain untouched until freezing.

***Evidence/files:** This protocol was later formalised in `data/raw/DATA_MANIFEST.md` and final report Sections 3.2-3.3, then implemented in `src/data/load.py`, `src/data/split.py`, `src/data/preprocess.py`, and `src/data/resample.py`.*

Challenges Encountered (if any): Leakage could occur through duplicate transactions, scaling, resampling, autoencoder fitting, threshold selection, or unsafe joins between predictions and explanations.

Solutions / Actions Taken (if any): I introduced content deduplication, stable case IDs, train-only preprocessing and resampling, validation-only selection, and manifest-based checks.

Progress Reflection: Stable `case_id` values became necessary because SHAP, LLM, and application records should never be joined by row position. I also recognised that leakage prevention had to be demonstrated by tests and manifests, not only described in the report.

Plans for Next Week: Finalise the Autoencoder and XGBoost component design.

Supervisor's Feedback (if any): —

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Week Number:	6
Date Range:	15-21 June 2026

Week 6: Detector Architecture and Hybrid-Feature Plan

Objectives for the Week: Translate the model proposal into components that could be compared under one interface.

Activities Completed: I planned a common XGBoost wrapper, validation-only early stopping, validation-selected thresholding, an Autoencoder trained on legitimate training observations, reconstruction-error features, and latent features. SMOTE and class weighting were kept as separate imbalance strategies so their effects would not be mixed.

***Evidence/files:** The component design was later formalised in final report Sections 3.4-3.6 and implemented in `src/models/xgb.py`, `src/models/autoencoder.py`, `src/evaluation/metrics.py`, and `src/evaluation/threshold.py`.*

Challenges Encountered (if any): The original title could encourage selecting the hybrid model regardless of the result. I instead planned all groups to use the same split, metrics, artifact contract, and seeds.

Solutions / Actions Taken (if any): I placed every detector group behind the same XGBoost interface, split, seeds, metrics, and artifact contract instead of favouring the hybrid configuration.

Progress Reflection: A negative Autoencoder result would still answer a valid research question. This was more academically useful than changing the metric or comparison after seeing the outcome.

Plans for Next Week: Define the SHAP-to-LLM evidence boundary and narrative failure checks.

Supervisor's Feedback (if any): —

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Week Number:	7
Date Range:	22-28 June 2026

Week 7: Explainability, Local LLM, and Guardrail Design

Objectives for the Week: Specify how generated text would remain subordinate to detector evidence.

Activities Completed: I designed signed SHAP reason codes containing feature identity, rank, and contribution direction. The local Ollama model would receive only a minimized reason-code package, not a raw transaction row, exact values, labels, probabilities, or SHAP magnitudes. Candidate narratives would be checked for format, completeness, grounding, and direction. A failed check or unavailable LLM would return a deterministic brief.

***Evidence/files:** This boundary was later formalised in final report Sections 3.6-3.8 and implemented in `src/explainability/`, `src/narratives/`, and `corpus/guardrail_corpus_v1.jsonl`.*

Challenges Encountered (if any): Automatically normalizing or repairing rejected LLM text would make the output look correct while hiding what the model originally generated.

Solutions / Actions Taken (if any): I separated raw-output measurement from delivered-output policy and rejected repair or normalisation of failed LLM text.

Progress Reflection: I separated OFF-policy measurement of raw output from ON-policy delivery. This allowed the project to report the LLM's observed failure rate while ensuring that detected failures were not shown as accepted evidence.

Plans for Next Week: Design an application that demonstrates the same evaluated evidence chain.

Supervisor's Feedback (if any): —

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Week Number:	8
Date Range:	29 June-5 July 2026

Week 8: Analyst Workbench Architecture

Objectives for the Week: Plan a deliverable application that resembles an analyst workflow rather than a model demonstration page.

Activities Completed: I selected React and TypeScript for the frontend and FastAPI for the local backend. The planned workflow used one alert queue, a case review workspace, explanation assurance, evaluation results, and separate analyst actions. Research artifacts would be immutable consumers, while status, notes, and activity would be stored in a separate local SQLite workflow plane.

Evidence/files: This architecture was later formalised in docs/specs/2026-07-13-react-fastapi-demo-dashboard-spec.md and final report Sections 3.10-3.11, then implemented in app/frontend/ and app/backend/.

Challenges Encountered (if any): A dashboard organised around tabs for models and charts could appear fragmented and would not explain who uses the system or what action follows an alert.

Solutions / Actions Taken (if any): I organised the application around one analyst workflow: select an alert, inspect evidence, review validation, and record an action.

Progress Reflection: The application was reframed around the sequence “select alert, inspect evidence, review validation, record an action.” It would support decision-making but would not declare whether a transaction was truly fraudulent.

Plans for Next Week: Prepare the implementation contracts, provenance schema, and review criteria.

Supervisor's Feedback (if any): —

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Week Number:	9
Date Range:	6-12 July 2026

Week 9: Implementation Readiness and Evidence Contracts

Objectives for the Week: Convert the design into an executable build sequence with clear artifact boundaries.

Activities Completed: I prepared the implementation work breakdown for data, models, experiments, SHAP, narrative generation, provenance, results, and the application. I defined run manifests, stable source-run links, frozen artifacts, and adversarial review expectations. Tests were planned where they protected leakage, provenance, validation, and workflow contracts, rather than forcing every documentation or presentation task into TDD.

***Evidence/files:** This readiness work was formalised at the start of Week 10 in docs/plans/2026-07-13-cp2-implementation-plan.md, docs/specs/2026-07-13-react-fastapi-demo-dashboard-spec.md, and docs/specs/2026-07-14-fraud-review-workbench-spec.md.*

Challenges Encountered (if any): The system had many interfaces. If detector predictions, SHAP records, narratives, and dashboard cases used inconsistent identities or unverified "latest" directories, the final demonstration could silently show a different system from the reported experiment.

Solutions / Actions Taken (if any): I defined run manifests, stable source-run links, frozen artifact paths, and targeted tests at leakage, provenance, validation, and workflow boundaries.

Progress Reflection: Treating the plan and manifests as contracts made later results easier to audit. The most important tests were placed at research-risk boundaries instead of using TDD mechanically for every task.

Plans for Next Week: Implement and execute the full detector-to-explanation pipeline.

Supervisor's Feedback (if any): —

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Week Number:	10
Date Range:	13-19 July 2026

Week 10: Core Implementation, Detector Benchmark, and First Review Package

Objectives for the Week: Build the reproducible research pipeline and obtain the first complete experimental evidence.

Activities Completed: I implemented dataset integrity checks, stable case IDs, preprocessing, SMOTE, the XGBoost and Autoencoder components, thresholding, manifests, and leakage audits. I ran six groups over five seeds, producing 30 detector runs. G6 had the numerically highest mean test AP at 0.855214, although the leading differences were small and the Autoencoder variants showed no clear advantage. G6 seed 42 was frozen for SHAP. I generated reason codes for 51 flagged cases, ran the strict and simple Ollama prompt arms, calibrated the validator, implemented the first React-FastAPI workbench, and prepared a supervisor review package.

Evidence/files: *Git commits from dc19eb8 onward; experiments/runs/; reports/tables/results_summary.csv; experiments/runs/2026-07-14_g4_seed42/; experiments/runs/2026-07-14_g5_seed42/; app/; supervisor_meeting/.*

Challenges Encountered (if any): The original Autoencoder claim was not supported. The ULB feature names were also anonymous, so LLM narratives remained close to the SHAP reason-code wording.

Solutions / Actions Taken (if any): I retained the negative Autoencoder result, froze G6 seed 42 for downstream explanation work, and used the first workbench review to identify communication gaps.

Progress Reflection: I kept the negative detector result and repositioned the project around verifiable explanation delivery. The first workbench made the evidence visible but still needed a clearer operational story and better result visualisation.

Plans for Next Week: Present the progress, respond to feedback, and strengthen the readable explanation evaluation.

Supervisor's Feedback (if any): —

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Week Number:	11
Date Range:	20-26 July 2026

Week 11: Supervisor Feedback, Result Visualisation, and S0 Evaluation

Objectives for the Week: Improve how the project communicates detector trade-offs and demonstrate the value and limitations of the local LLM more clearly.

Activities Completed: During the supervisor discussion, the main practical feedback was to show more bar charts, especially F1, and to make the purpose of the website and the value of Ollama clearer. I added mean F1 with standard- deviation error bars, grouped precision and recall, false-positive and false-negative counts, and narrative violation, fallback, and acceptance charts. I also ran an exploratory model search but retained the frozen G6 chain because the alternatives did not justify breaking provenance. To address the anonymous ULB limitation, I added the S0 synthetic stream with readable transaction concepts. Its frozen test result was AP 0.544017 and F1 0.514286. Among 25 selected alerts, 23 guarded LLM briefs passed and 2 used fallback.

Evidence/files: docs/reviews/2026-07-20-detector-model-search.md; reports/figures/detector_metric_bars.png; reports/figures/narrative_delivery_bars.png; docs/specs/2026-07-26-semantic-fraud-triage-extension.md; experiments/runs/2026-07-26_s0_seed42/; reports/semantic_results_manifest.json.

Challenges Encountered (if any): S0 could make the project appear to contain two unrelated detectors. The local LLM also did not consistently produce a more complete brief than the deterministic renderer.

Solutions / Actions Taken (if any): I added the requested result bar charts, retained the frozen detector after exploratory search, and positioned S0 as a readable semantic context rather than a competing benchmark.

Progress Reflection: I defined ULB as the supporting real-data detector benchmark and S0 as the primary readable semantic and operational context. The LLM's value is controlled articulation and measurable failure handling, not new risk knowledge. The accepted S0 outputs choosing shorter summaries was retained as a negative result.

Plans for Next Week: Complete the human pilot, final report, presentation, and submission verification.

Supervisor's Feedback (if any): Show more bar charts, especially for F1, and clarify the purpose of the website and the value of Ollama.

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Week Number:	12
Date Range:	27-31 July 2026

Week 12: Human Pilot, Final Report, Presentation, and Submission Preparation

Objectives for the Week: Integrate the completed evidence into the assessed CP2 deliverables before the 31 July deadline.

Activities Completed: I prepared the consent material, questionnaire, recruitment message, analysis plan, and explanation stimuli. Eleven adult proxy reviewers completed 99 case reviews. Because this was below the planned minimum of 18 and target of 30, I reported it as descriptive interim evidence and did not run a superiority test. I also manually audited all 49 strict-arm narratives delivered without fallback against their bound evidence and found no semantic violation; the 95% Wilson interval was 0%-7.27%, so I retained the single-reviewer and uncertainty limits. I completed the rubric-aligned final report, 12-week logbook, 12-slide presentation, presenter guide, submission index, rubric matrix, page-map verification, and software test record. As of 29 July, the final report was 79 A4 pages and the tested application remained a local prototype rather than a production-bank deployment.

Evidence/files: docs/studies/; reports/human_eval_results.json; reports/tables/human_eval_*.csv; reports/thesis/submission/; reports/presentation/; reports/RUBRIC_ALIGNMENT_MATRIX.md; reports/thesis/submission/VERIFICATION_RECORD.md; experiments/audit/audit_result.json.

Challenges Encountered (if any): The human sample was smaller than planned, and the submission still required consistency checks across the report, logbook, slides, figures, page numbers, and stated limitations.

Solutions / Actions Taken (if any): I completed the human pilot analysis, 49-item manual audit, report, logbook, presentation, verification records, and submission package without overstating the evidence.

Progress Reflection: The final package is strongest when it reports both positive and negative evidence. The project demonstrates a reproducible detector-to-explanation chain and a fail-closed local-LLM delivery boundary, but the 0/49 manual-audit observation still does not prove universal semantic correctness. It also does not prove a new best detector, improved analyst productivity, or production readiness.

Plans for Next Week: Submit the final report and logbook by the course deadline, rehearse the system demonstration and viva, and continue recruitment only if additional responses can be incorporated without changing the frozen evidence claims improperly.

Supervisor's Feedback (if any): —

Supervisor's Signature

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